A Predictive Analysis of Café Sales Data Using Linear Regression and Attribute Selection

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***Abstract*— Linear Regression is a fundamental statistical and machine learning technique widely used for predictive modeling due to its simplicity, interpretability, and effectiveness in identifying relationships between variables. This study investigates the performance of simple and multiple linear regression models using the Dirty Café Sales dataset obtained from Kaggle, which simulates real-world data challenges through missing values, inconsistencies, and outliers. The objective of this work is to develop accurate predictive models for total sales spending by applying a complete end-to-end predictive modeling pipeline. The methodology includes data acquisition, exploratory data analysis, data cleaning and preparation, correlation-based attribute selection, model construction, performance evaluation, and visualization. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Experimental results demonstrate that while simple linear regression provides a meaningful baseline, multiple linear regression significantly improves predictive accuracy through the inclusion of additional relevant predictors. This study highlights the importance of data preprocessing and attribute selection in building robust predictive models and discusses practical considerations when applying linear regression to real-world business data.**

Keywords—Linear regression, Predictive Modeling, Attribute Selection, Data Preprocessing, Regression Analysis

I. Introduction

Many businesses have undergone this process, creating linear regression models to uncover patterns in data and support informed decision making. As organizations increasingly rely on data-driven strategies, the ability to develop accurate and reliable predictive models has become essential. This exercise aims to provide hands-on experience with the end-to-end predictive modeling pipeline, including data cleaning, variable selection, and model training and testing. These procedures are essential for turning unstructured data into meaningful insight and ensuring the accuracy of predicted outcomes.

Data cleaning and preparation plays a crucial role in ensuring that the machine learning (ML) models can effectively learn from the data. By addressing the issues such as missing values (NaN), inconsistent categorical labels, and unscaled numerical features, the prediction of the model can leverage or improve its performance.

Additionally, Linear Regression is one of the most widely used techniques in predictive modeling due to its simplicity, interpretability, and effectiveness in modeling relationships between dependent variables (target) and independent variables (predictors). However, the performance of linear regression depends on the quality pipeline and the selection of relevant attributes. Attribute selection techniques are methods that play a crucial role in improving model accuracy by reducing the available attributes and focusing only on those that contribute meaningful information. It also helps prevent the model from learning noise and minimizing overfitting, and enhancing interpretability to easily understand and explain the correlation of predictors to target variables. Through this exercise, linear regression and attribute selections go hand-in-hand to examine the highlight of developing robust and reliable predictive models for real-world applications.

II. Data Collection

A. Dataset Overview

The dataset that was used in this activity is the dirty_cafe_sales from the kaggle website. This dataset contains transactional sales records from a cafe and it is intentionally designed with inconsistencies, missing values, invalid entries, duplicate records, and outliers to simulate real-world scenarios.

Dataset exploration was conducted as a first step to understand the structure, variables, and data types. The analysis revealed that the dataset consists of both numerical such as Quantity, Price Per Unit, Total Spent, and Dates, and categorical attributes, including product information, transactional details, and sales-related variables. This understanding was essential as it determined the appropriate data cleaning and preprocessing techniques that will be used later.

B. Process of Cleaning the Data

To address the identified data quality issues, a structured data cleaning and preparation pipeline was implemented using pandas library. Irrelevant attributes such as transaction ID were removed, as they do not contribute to predictive modeling. Placeholder values representing missing or

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erroneous data such as ERROR, UNKNOWN, N/A, null, NaN values were standardized and converted into uniform missing values. Numerical attributes, including the Quantity, Price Per Unit, and Total Spent, were cleaned by removing invalid text values and converting them into numeric format.

A rule-based strategy was applied to handle missing numerical values by leveraging known relationships within the dataset. Missing values for Total Spent, Quantity, or Price Per Unit were estimated using the relationship Total Spent = Quantity x Price Per Unit when sufficient information was available. Remaining values were computed using the median of each numerical attribute to reduce the influence of bigger values. Moreover, quantity values were rounded and converted to integer format to maintain logical consistency.

On the other hand, the categorical attributes such as Item, Payment Method, and Location were cleaned by standardizing text formatting and replacing missing or invalid entries with a uniform "Unknown" label. Transaction dates were converted into date time format, and missing values were imputed using the most frequently occurring date. Features including the transaction month, date, and day of the week, were then extracted to enhance predictive capability, after which the original date attribute was removed.

Moreover, duplicated records were identified and removed to prevent bias during model training. Outliers in the target variable (Total Spent_ were detected using the interquartile range (IQR) method and capped through winsorization to reduce their influence without discarding valid observations. Finally, categorical variables were encoded using one-hot encoding, and some remaining attributes that are irrelevant were excluded to simplify the modeling process. These data cleaning and preparation steps ensured that the dataset was consistent, reliable, and suitable for linear regression and attribute selection analysis.

III. Attribute Selection

A. Concept of Attribute Selection

Attribute selection is the process of identifying the most relevant variables that contribute to predicting a target variable. It is essential in building efficient predictive models because it reduces model complexity, minimizes overfitting, and improves interpretability. By selecting only relevant attributes, regression models can focus on meaningful relationships rather than noise which leads to an improvement in accurate prediction and more reliable results.

B. Correlation Analysis as an Attribute Selection

Correlation was used as an attribute selection technique alongside Heatmap and VIF, though VIF is not correlation, but it supports attribute selection. Both can be used to evaluate the linear relationship between predictor variables and target variable. Pearson correlation coefficients were computed among all numerical attributes after categorical variables were encoded into numerical form. The correlation values were then examined and ranked to determine which variables exhibited the strongest association with the target variable. Highly correlated attributes were identified as potential predictors for regression modeling, while variables with weak or negligible correlation were considered less informative. This analysis enabled the selection of the most influential features for both simple and multiple linear regression models, ensuring that the resulting models were both efficient and effective for predictive analysis.

```python
# CORRELATION
corr = df_model.select_dtypes(include=["number"]).corr()
target_corr = corr[target].sort_values(ascending=False)

print("\nCorrelation with Total Spent (Top 15):")
print(target_corr.head(15))

# BEST FEATURE FOR SIMPLE REGRESSION
best_feature = target_corr.index[1]
print("\nBest predictor for SIMPLE regression:", best_feature)
```

Fig. 1. Correlation Code

C. Results of Correlation Coefficients

The correlation coefficients were analyzed and ranked according to their strength in relation to the target variable, Total Spent. Attributes with higher absolute correlation values were identified as more influential predictors and selected for regression modeling. Among these, Quantity exhibited the strongest correlation with the target variable and was used as the predictor for simple linear regression analysis. Additional highly correlated attributes were incorporated into multiple linear regression models to further improve predictive performance. The results of the correlation analysis were also supported using a correlation heatmap and VIF, which visually illustrated the relationships among the numerical features.

```
Correlation with Total Spent (Top 15):
Total Spent       1.000000
Quantity          0.703219
Price Per Unit    0.644050
trans_day        -0.001317
trans_dow        -0.006017
trans_month      -0.007917
Name: Total Spent, dtype: float64

Best predictor for SIMPLE regression: Quantity
```

Fig.2. Identifying Predictor

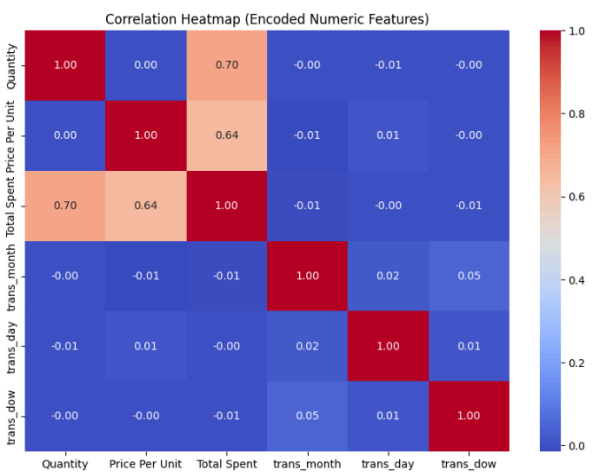

Fig.4. Heatmap Result

```
--- VIF Results (Top 15 highest) ---
          Feature       VIF
1  Price Per Unit  4.576792
0        Quantity  4.230301
2     trans_month  3.733304
3       trans_day  3.536853
4       trans_dow  2.853917
```

Fig. 3. VIF Results

IV. Model building: Simple and Multiple Linear Regression

A. Model Construction

Both simple and multiple linear regression models were developed using the highest correlation that was presented by the rankings, heatmap, and VIF variables.

For the simple linear regression model, the predictor that exhibited the highest correlation with the target variable, Total Spent, was Quantity. The dataset then was split into training and testing sets using an 80:20 ration to ensure unbiased model evaluation. The simple linear regression model was trained using the training data to estimate the relationship between the selected predictor and the target variable.

For Multiple linear regression, highly correlated attributes were also selected as predictors to capture more complex relationships within the dataset. These predictors were used to construct a multiple linear regression model following the same train-test split approach. Model coefficients were analyzed to assess the contribution of each predictor to the target variables, allowing for a comparison between the simple and multiple regression model.

B. Model Performance Evaluation and Interpretation

The performance of both regression models was evaluated using the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2) to assess prediction accuracy. For the simple linear regression modal, Quantity was used as the only predictor of Total Spent (target). The model achieved an MAE of 3.17 and an RMSE of 4.18, indicating a moderate average prediction error. The R^2 value of 0.50 suggests that approximately 50% of the variance in total spending can be explained by quantity alone. Moreover, The positive slope coefficient further confirms a direct linear relationship between quantity purchased and total spending. However, it only uses one predictor, so it cannot account for other important influences.

In contrast, the multiple linear regression model utilizes additional predictors, including Price Per Unit and transaction day, to account for pricing variability. This model achieved a significantly lower MAE of 1.27 and RMSE of 1.77, which reflect the improvement of prediction accuracy. The R^2 value of 0.91 indicates that over 90% of the variance in total spending is explained by the selected predictors, demonstrating an improvement in model fit. The regression coefficient reveals that both quantity and price per unit have strong positive contributions to total spending, while the transaction day is a minor influence.

TABLE 1. SIMPLE LINEAR REGRESSION PERFORMANCE METRICS

Metric	**Value**	**Description**
Predictor used	Quantity	Independent variable used to predict total spending
Slope	2.93	Total spending increases by approximately 2.93 units for each additional item purchased
Intercept	0.01	Baseline value for spending when quantity is 0
MAE	3.17	Average absolute prediction error
RMSE	4.18	Penalize larger prediction error
R^2	0.50	Indicates that 50% of the variance in total spending is explained by model

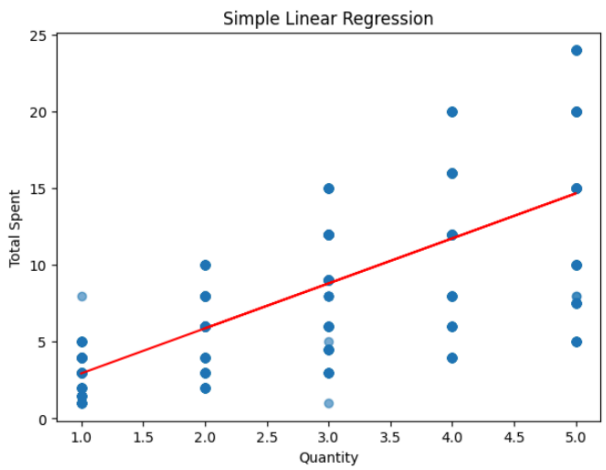

Fig. 6. Simple Linear Regression Performance Evaluation

TABLE 2. MULTIPLE LINEAR REGRESSION PERFORMANCE METRICS

Metric	Value	Description
Predictor used	Quantity, Price Per Unit, trans_day	Independent variable used to predict total spending
Quantity Coefficient	2.92	Indicates an Increase in total spending as quantity increases
Price Per Unit Coefficient	2.96	Reflects the impact of unit price on total spending
Trans_day Coefficient	-0.002	Indicated a minimal effect on total spending
Intercept	-8.62	Baseline value of total spending when all predictors are 0
MAE	1.27	Average absolute prediction error
RMSE	1.77	Penalize larger prediction error
R^2	0.91	Indicates that 91% of the variance in total spending is explained by model

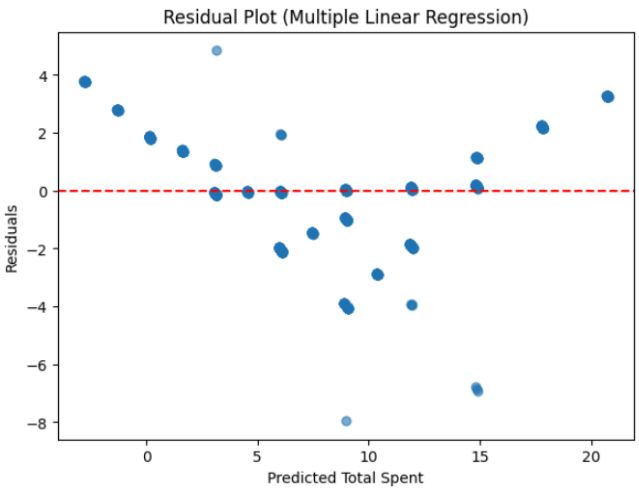

Fig. 7. Multiple Linear Regression Performance Evaluation

V. DISCUSSIONS

A. Model Construction

TABLE 3. COMPARISON OF SIMPLE AND MULTIPLE LINEAR REGRESSION PERFORMANCE

Model Type	Predictors Used	MAE	RMSE	R^2
Simple	Quantity	3.17	4.18	0.50
Multiple	Quantity, Price Per Unit, trans_day	1.27	1.77	0.91

The table clearly demonstrates a clear performance difference between simple and multiple linear regression models. The simple linear regression model, which uses Quantity as the predictor variable, provides a useful baseline by capturing the general linear relationship between quantity purchased and total spending. However, its moderate performance metrics, including a higher MAE and RMSE and and R^2 value of approximately 0.50, indicates that a significant amount in total spending remains unexplained. This limitation highlights the inability of a single predictor to fully represent the complex real-world purchasing behaviour.

On the other hand, the multiple linear regression model outperformed the simple model by incorporating additional relevant predictors such as Price Per Unit and transaction day. The inclusion of these attributes leads to a drastic reduction of prediction error and a significant increase in R^2 value of approximately 0.91. The improved performance indicates that the total spending is influenced by multiple factors, and having more of these relationships result in a more accurate and reliable predictive model.

B. Impact of Attribute Selection on Model Performance

The attribute selection process plays a crucial role in improving the model performance. By applying correlation analysis, we were able to identify the most influential predictors with the target variable. This selection ensured that only relevant attributes are included for regression models. Which reduces noise and improves interpretability. The selection of Quantity as the only predictor for simple regression established a meaningful baseline, while the inclusion of additional highly correlated attributes in multiple regression models enhanced predictive accuracy.

Furthermore, the use of heatmap and Variance Inflation Factor (VIF) helped evaluate the relationship among predictors and verify their suitability for regression analysis

VI. Conclusion and reflection

This activity demonstrated the complete end-to-end process of predictive modeling using linear regression, from data collection and preprocessing to attribute selection, model construction, and performance evaluation. The results showed that simple linear regression, using the Quantity as a single predictor, provides a meaningful baseline. However, the multiple linear regression model, which utilizes more relevant predictors such as Price Per Unit and trans_day, significantly improved the predictive performance and findings, achieving a lower error and higher coefficient. These findings emphasize the importance of proper data preparation and attribute selection in creating accurate and reliable predictive models.

There are several challenges I have encountered during the activity. One major challenge was identifying an appropriate dirty dataset, as it is necessary to inspect and explore the dataset at an early stage to determine whether the data was suitable for regression analysis. This inspection helped me identify which issues such as missing values, inconsistencies, and meaningful relationship between variables I have to provide solutions. Moreover, another challenge I have encountered is constructing a complete data cleaning pipeline, since it is crucial to reduce noise, incompleteness, inconsistency, and outliers resulting in a smooth flow of integrating regression.

These challenges I have faced were addressed by revising the previous laboratory activities and pdf lesson posted on Teams to refresh all concepts and by using AI-assisted explanations to better understand the functionality of each preprocessing and modeling. Overall, this activity taught me techniques and valuable experience in handling real-world data for future predictive modeling tasks.